# Structural Defects Network: MLOps project using ResNet-18 based binary image classifier

**Course ID.: CPE393 Machine Learning Operations**


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# ABSTRACT

*TBD...Brief summary of the project: objective, method, dataset, key results and potential impact...*

*200 - 250 words*

- *Problem Statement (Structural Defects Network)*
- *Data Collection (Kaggle, ... Any other data source)*
- *Train/Test (80-20, 70-30, 60-40)*
- *Evaluation Metrics (Optuna tuning), F1 metrics*
- *Results (e.g. We achieved 75% F1 score on val dataset with ResNet-18)*
- *Fast API, Containerized Model*


**Keywords**: *Coming soon...*

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# CHAPTER 1

# INTRODUCTION

## 1.1 Problem Statement

As the recent earthquake in Thailand caused significant structural damage, particularly to high-rise buildings which led to widespread concrete cracking and breakage. [1] Therefore, the need for efficient and reliable damage assessment has become more critical than ever. Structural defects such as cracks and spalling not only undermine the integrity of infrastructure but also pose serious risks to public safety if not detected and repaired promptly.

Traditional inspection methods rely heavily on manual labor which are time-consuming and are prone to human error making them impractical for large scale assessments especially in post-disaster scenarios. By that we want to enhance the current method by using modern technology from this project instead. [2]

This project proposes an AI-powered defect detection system that leverages deep learning to automate the identification and localization of structural damage from images. By utilizing pre-trained models such as ResNet [3], EfficientNet[4] and MobileNetV3[5]. The system classifies images as either cracked or non-cracked and applies object detection techniques for precise localization. This approach not only improves the speed and accuracy of structural assessments but also supports engineers and decision-makers in prioritizing repairs, ultimately enhancing infrastructure resilience after natural disasters.

## 1.2 Project Scope and Goals

The goal of this project is to enhance structural health monitoring through automated crack detection, reducing manual inspection and improving safety. To achieve this, the project will:

1. Develop a classification model (ResNet, EfficientNet and MobileNetV3) in order to distinguish between cracked and non-cracked surfaces with high accuracy.

2. Implement object detection and segmentation using Faster R-CNN for precise localization of cracks.

3. Preprocess and enhance data, including image augmentation and feature extraction for improved model performance.

4. Train and evaluate models using relevant performance metrics such as accuracy, precision, recall, F1-score, AUC-ROC and Confusion Matrix.

5. Deploy the best-performing model via Flask/FastAPI, Docker and cloud platforms (AWS, GCP, Azure or Hugging Face Spaces) for real-world applications.

6. Explore optional CI/CD integration, including automated model retraining, performance monitoring, and deployment pipelines.

This project will contribute to efficient and scalable crack detection, improving infrastructure maintenance and safety through AI-driven automation.

## 1.3 Data Description

## 1.4 Keywords

CHAPTER 2

# IMPLEMENTATION

## 2.1 Data Pre-processing

### 2.1.a Handling Class Imbalance

As the original SDNET dataset exhibits significant class imbalance, with non-cracked images comprising approximately 85% of the data. To mitigate this issue and improve model sensitivity to cracks, two resampling strategies were applied:

**Balanced 50/50 dataset** Random undersampling matched the number of non-cracked images to the number of cracked images. This ensures equal representation and helps prevent model bias.

**Balanced 40/60 dataset** A less aggressive undersampling approach was used, setting a cracked-to-non-cracked ratio of 40:60. This retains more samples while reducing imbalance.

These resampled datasets allow for performance comparisons under different balance conditions, ensuring the model's robustness across varying class distributions.

### 2.1.b Image Preprocessing

Before being fed into the model, images underwent several preprocessing steps aimed at enhancing their quality and diversity:

**Resizing** All images were resized to the same resolution for uniformity.

**Denoising** A mild blur was applied to reduce noise while preserving important features such as crack edges.

**Normalization** Pixel values were scaled to a standardized range to help the model learn more effectively.

These steps ensured that the input data was clean, consistent, and suitable for deep learning.

### 2.1.c Data Augmentation

To enhance the robustness of the model and prevent overfitting, data augmentation techniques were applied to the training image, which included:

**Random horizontal flipping** to simulate mirrored crack patterns.

**Small-angle rotation** $\pm 15°$ to mimic slight variations in camera angle.

**Brightness and contrast adjustments** to reflect differing lighting environments.

These augmentations simulate variations in lighting and camera angles, helping the model generalize better to new, unseen data.

### 2.1.d Dataset Construction and Splitting

### 2.1.e EDA (Exploratory Data Analysis) findings

### 2.1.f Class Distribution by Surface Type

### 2.1.g Class Distribution Across Datasets

### 2.1.h Sample Images Overview

CHAPTER 3

# Model Architecture

## 3.1 Classification Models

### 3.1.a Custom CNN (Convolutional Neural Network)

### 3.1.b ResNet (Residual Network)

[3]

### 3.1.c MobileNetv3

[6][5]

## 3.2 Object Detection Models

### 3.2.a Faster R-CNN

[7]

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